Software Requirements Specification

for

AI-Powered

Tour Guide Booking System

with Voice and RAG

Version 1.0 approved

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SLIIT

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Table of Contents

Table of Contents **2**
Revision History **2**
1. Introduction **1**
1.1 Purpose 1
1.2 Document Conventions 1
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 2
1.5 References 3
2. Overall Description **4**
2.1 Product Perspective 4
2.2 Product Functions 5
2.3 User Classes and Characteristics 7
2.4 Operating Environment 9
2.5 Design and Implementation Constraints 9
2.6 User Documentation 11
2.7 Assumptions and Dependencies 11
3. External Interface Requirements **12**
3.1 User Interfaces 12
3.2 Hardware Interfaces 17
3.3 Software Interfaces 17
3.4 Communications Interfaces 18
4. System Features **19**
4.1 Impact Analysis ***20***
5. Other Nonfunctional Requirements **24**
5.1 Performance Requirements 24
5.2 Safety Requirements 25
5.3 Security Requirements 26
5.4 Software Quality Attributes 27
5.5 Business Rules 27
6. Other Requirements **28**
Appendix B: Analysis Models **30**
Appendix C: To Be Determined List **32**

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed overview that defines functional and non-functional requirements for the AI-Powered Tour Guide Booking System with Voice and RAG. The system connects tourists and local guides, providing voice-enabled queries, secure guide verification, RAG-based knowledge retrieval, personal recommendations, booking flows, and emergency support. The SRS is the basis for design, implementation, testing and validation.

1.2 Document Conventions

IEEE standards are followed to make this document to maintain good quality and higher standardization.

- The font style of the whole document is Times New Roman.
- The font size of the text body is 12.
- The font size of the main headings is 18.
- The font size of the subheadings is 14.
- All the main and subheadings are bolded.
- All the subheadings are numbered with the floating-point numbers.

1.3 Intended Audience and Reading Suggestions

The document is intended for the following audience.

- Developers and Engineers: To understand the system's requirements and technical details.
- Project Managers: To track milestones and ensure alignment with goals.
- Stakeholders and Clients: To validate that the system meets user needs.
- Testers: To design test cases based on specified requirements.
- End Users and Trainers: To comprehend system functionalities.

1.4 Product Scope

The proposed system is a secure, scalable AI-powered web application designed to connect tourists with qualified local tour guides through intelligent, privacy aware automation. The platform enhances the tourism experience in Sri Lanka by combining voice-based interaction, Retrieval-Augmented Generation (RAG), and AI-driven guide matching within a responsive web and API ecosystem.

The software enables tourists to submit voice or text queries in English, Sinhala, or Tamil, automatically extracting highlighted keywords and retrieving contextual information—such as attractions, safety advisories, and verified guide profiles from a vector-based knowledge database. Using this context, the system's AI engine recommends suitable guides and generates personalized travel insights.

Guides can register securely by uploading multiple verification documents, while privacy mechanisms ensure that personal details are locked and only revealed post-booking. Integrated modules handle multilingual chat agent assistance, real-time bookings, emergency alerts, and post-trip feedback, creating a transparent and trusted service network.

Purpose and Benefits:

- Improves trust and transparency in guide hiring through verified, tiered authentication.
- Saves time and effort for tourists by automating information retrieval and guide matching.
- Promotes inclusivity through multilingual voice interaction.
- Reduces fraud and enhances safety via AI-assisted verification and alerts.

This software supports digital tourism transformation and AI integration strategies aimed at promoting Sri Lanka as a smart, secure, and user-centric travel destination advancing both public-sector modernization and private-sector competitiveness within the tourism economy

1.5 References

[1] X. Zhou, Z. Zhou and X. Shi, "FCH-TTS: Fast, Controllable and High-quality Non Autoregressive Text-to-Speech Synthesis," 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 2022, pp. 1-8.

[2] S. Mohanty and B. K. Swain, "S2S system for voice oriented tourism information delivery in Indian context," 2013 International Conference Oriental COCOSDA held jointly with 2013 Conference on Asian Spoken Language Research and Evaluation (O-COCOSDA/CASLRE), Gurgaon, India, 2013, pp. 1-5.

[3] G. Boza-Quispe, J. Montalvan-Figueroa, J. Rosales-Huamaní and F. Puente-Mansilla, "A friendly speech user interface based on Google cloud platform to access a tourism semantic website," 2017 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON), Pucon, Chile, 2017, pp. 1-6
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[4] Y. Wang, J. Niu, X. Deng and H. Li, "Gfl-TTS: A text-to-speech model that combines new tonal prediction and alignment," 2024 4th International Conference on Neural Networks, Information and Communication Engineering (NNICE), Changchun, China, 2024, pp. 1-6.

[5] D. O'Shaughnessy, "Timing patterns in fluent and disfluent spontaneous speech," 1995 International Conference on Acoustics, Speech, and Signal Processing, Detroit, MI, USA, 1995, pp. 240-243.

2. Overall Description

2.1 Product Perspective

The proposed software is a new, self-contained web-based intelligent tourism platform designed to digitalize and personalize the tour guide booking experience. It is not a follow-on product from any existing application but rather an innovative system that integrates multiple emerging technologies including Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and voice-based interaction — into one unified solution for Sri Lanka's tourism ecosystem.[1]

The system operates as a centralized AI-powered platform that connects tourists and registered guides. It integrates seamlessly with several external systems and APIs to provide real-time, context-aware services:

- Vector Database stores embedded knowledge documents for fast retrieval.
- LLM Provider (OpenAI GPT, Llama 2, or Mistral) generates contextually accurate and conversational responses.
- Embedding Models (OpenAI or Hugging Face) transform text documents into vector representations.
- Verification & Identity APIs authenticate guide documents and police checks.
- Payment Gateways handle secure financial transactions (optional).

These external integrations are orchestrated through an AI Agent Layer (LangChain or Llama Index), which coordinates data retrieval, processing, and output generation for user interactions.

2.2 Product Functions

The AI-Powered Tour Guide Booking System with Voice and RAG integrates intelligent automation, voice interaction, and AI-assisted verification to enhance the trust, convenience, and personalization of tourism experiences. Its major functional modules include the following:

2.2.1 Tourist Guide Interaction and Booking Management

This module facilitates seamless communication and transactions between tourists and registered tour guides through both voice and text-based interfaces. Tourists can search for guides by language, location, specialization, and price range using natural language queries. The system displays comprehensive guide profiles, including ratings, reviews, and verification status (verified or non-verified).[2]
It supports full booking workflows, allowing tourists to request, confirm, or cancel bookings directly through the web interface or AI chat agent. Integration ensures secure transactions. After completing a trip, tourists can submit feedback and ratings, which update guide trust scores and influence future recommendations. This function ensures a smooth, multilingual, and privacy-preserving experience for both tourists and guides.

2.2.2 AI Agent and RAG-Based Knowledge Retrieval

The AI Agent forms the intelligent core of the system, enabling context-aware assistance and real-time information retrieval. When a tourist submits a voice query, the system converts speech to text, identifies key entities and highlighted keywords, and retrieves related information such as nearby attractions, cultural events, safety guidelines, and available guides from a vector database integrating Retrieval-Augmented Generation (RAG), the system combines retrieved documents with Large Language Models (LLMs) to generate accurate, personalized, and multilingual responses.[3]

This ensures that tourists receive up to date, location-specific, and culturally relevant recommendations, improving the overall quality of decision-making.

2.2.3 Verification, Fraud Detection, and Privacy Control

This module ensures the authenticity and trustworthiness of tour guides and protects user data privacy. Guides upload multiple verification documents, including national ID, tourism department license, and police clearance certificates. The system uses OCR and AI-based validation to analyze and verify document authenticity, automatically classifying guides into verified or non-verified.
An integrated monitors booking patterns and user activities to identify anomalies, ensuring safer transactions. [4][5]

The system also implements, where sensitive personal details of both tourists and guides remain locked until booking confirmation. This layered verification and privacy mechanism enhances platform reliability and user confidence.

2.3 User Classes and Characteristics

The AI-Powered Tour Guide Booking System with Voice and RAG is designed to serve multiple categories of users, each with distinct roles, privileges, and interaction levels. The primary user classes are Tourists and Guides, while Administrators, System Integrators, and Third-Party Services play supporting or maintenance roles.

2.3.1 Tourist

Tourists are end-users seeking to find and book verified or non-verified tour guides through voice or text-based interaction. They form the largest and most frequent user group in the system.

- Typically, non-technical users require a simple and intuitive interface.
- Prefer multilingual support (English, Sinhala, Tamil) for accessibility.
- Expect privacy and data protection for personal information.
- Use mobile or desktop browsers to search, book, and communicate with guides.
- Rely on AI assistance for destination suggestions, price comparisons, and safety tips.

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2.3.2 Tour Guide

Registered service providers (local or professional guides) who advertise, verify, and manage their guiding services through the platform.

- Moderate technical literacy often requires guided onboarding.
- Must upload and verify documents (ID, licenses, police checks).
- Maintain dynamic availability, pricing, and location details.
- Value trust indicators (verification badges, ratings) to attract clients.
- Require secure handling of personal and financial data.

2.3.3 Admin and Verifier

Administrators and verifiers manage back-office operations, including account verification, fraud detection, and dispute resolution.

- Technically skilled staff with system-level privileges.
- Responsible for monitoring verification workflows, reviewing flagged accounts, and ensuring compliance with tourism regulations.
- May include tourism board officers or authorized platform managers.

2.3.4 System Integrators

Technical personnel responsible for system setup, deployment, updates, monitoring, and performance optimization.

- High technical expertise in server configuration, database management, vector DB orchestration, and AI integration.
- Infrequent direct interaction with the user interface; focus on backend maintenance and reliability.

2.4 Operating Environment

The AI-Powered Tour Guide Booking System with Voice and RAG operates within a modern, cloud-based environment designed for high scalability, cross-platform accessibility, and seamless AI integration. The system does not limit the usage of the functions according to the operating systems, allowing users to access it seamlessly on platforms such as Windows, macOS, Linux, Android, and iOS. Reachability of the system is maximized though the broad compatibility of the system across devices, ensuring accessibility to the target audience. Being a web-based system, there is no requirement to install any specific software for the functionality of the system.

A stable internet connection, geolocation features and a standard web browser are the primary requirements to use the system. The primary requirement is a stable internet connection and a standard web browser. The system is optimized for cross-platform usability, ensuring that users experience consistent functionality and performance across devices and browsers. By maintaining a device- and platform agnostic approach, the booking management system ensures accessibility, convenience, and ease of use for all potential users, contributing to its goal of promoting tour guide booking system.

2.5 Design and Implementation Constraints

The design and implementation of the AI-Powered Tour Guide Booking System with Voice and RAG are subject to several technical, regulatory, and operational constraints that define the development boundaries and influence architectural decisions. The system must comply with data privacy regulations, security standards, and integration requirements while maintaining performance and scalability within available resources follows.

Technical Stack Constraints	The frontend must be developed in React.js. and backend must be implemented using Node.js/Express.js
Integration Constraints	The system must seamlessly integrate with third-party APIs for LLM.speech-to-text, mapping, and payments.

Design and Coding Standards	Development must follow established secure SDLC practices, modular coding conventions, and consistent API versioning. All components should maintain interoperability and code readability for future maintenance.
Storage and Data Constraints	The system must efficiently handle large datasets, including vector embeddings and uploaded documents, within available cloud storage limits.

2.6 User Documentation

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Since user-friendliness is a top priority in the design of the system, significant external documentation is not necessary. To achieve this, the system integrates in-app help features that are set up strategically throughout the UI to provide contextually aware instructions. These are composed of brief explanations for important features, tooltips, and on-screen notifications. Users may quickly learn how to use the system without referring to outside resources because support is included right into the program.

This objective is further facilitated by the user interface's clarity and simplicity. Users can easily understand and engage with the system because of the design's straightforward and self-explanatory layouts, icons, and navigation routes. Consistent language and visual components improve usability and lessen the psychological workload on users. The system also follows best practices by using consistent formatting throughout, responsive design for device compatibility, and concise language. Together, these components ensure that every user has a seamless and convenient experience with the system. Users may concentrate on using the system to effectively meet their needs since traditional user manuals are no longer necessary thanks to the integration of in-app help and a carefully designed interface.

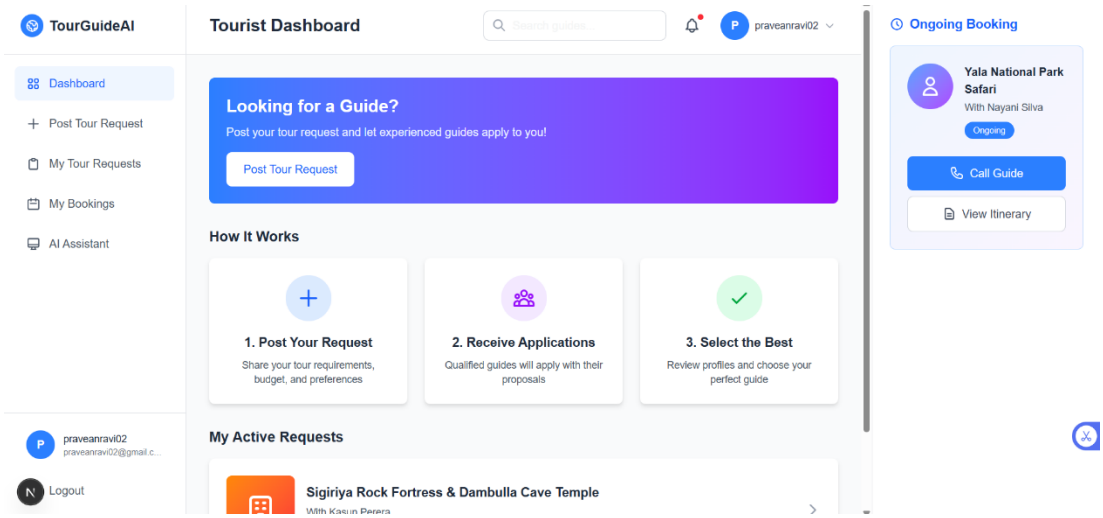
2.7 Assumptions and Dependencies

- The users of the system should have a device with the minimum hardware requirement to access the system such as mobile phone, laptop or desktop computer.
- Having the basic knowledge of technical device usage is expected.
- A stable internet connection.

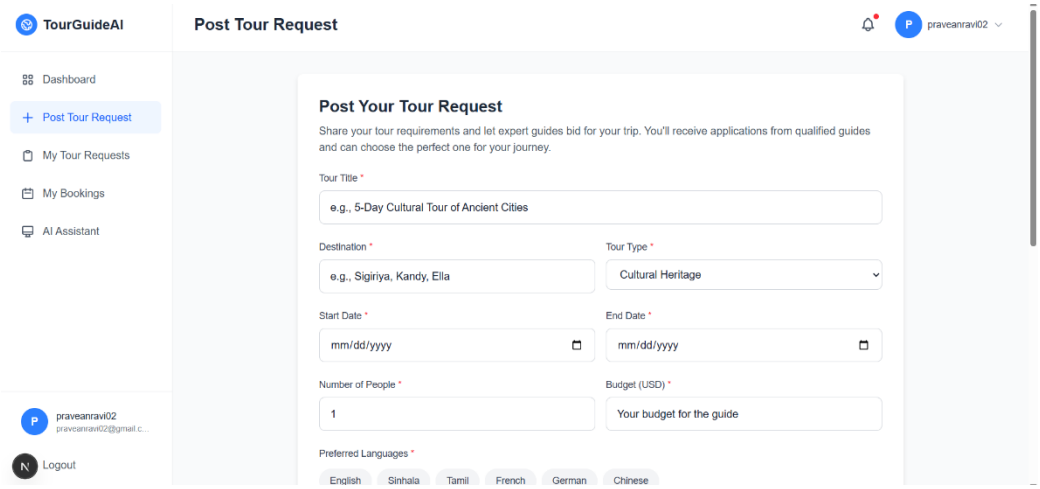
3. External Interface Requirements

3.1 User Interfaces

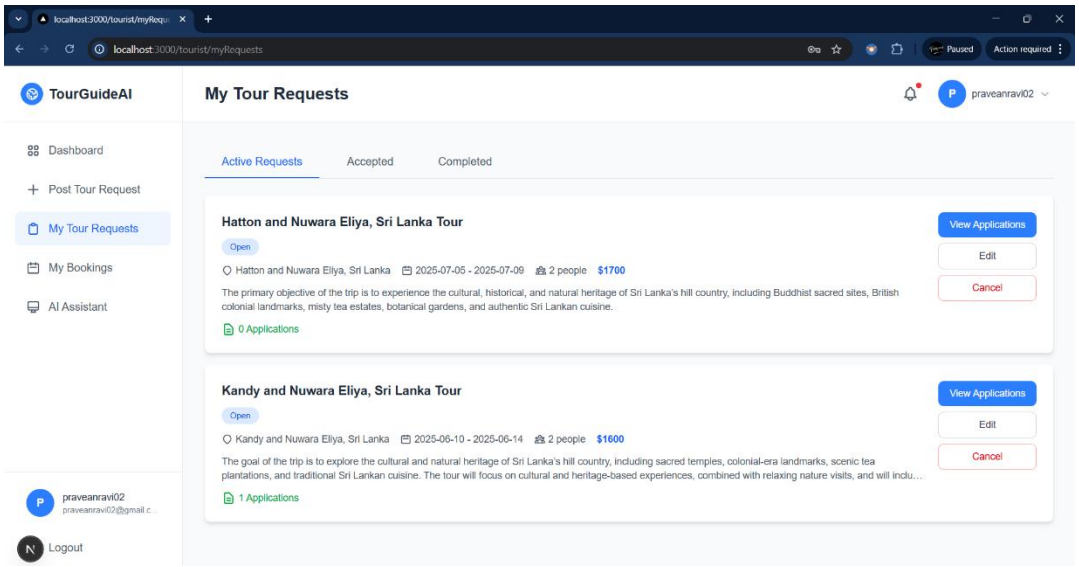
1. Tourist Home Page



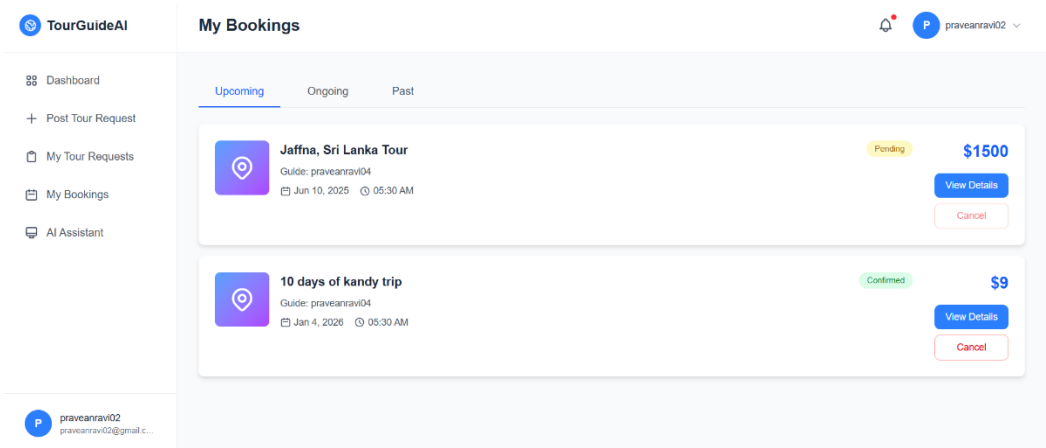
2. Tourist Post Tour Request



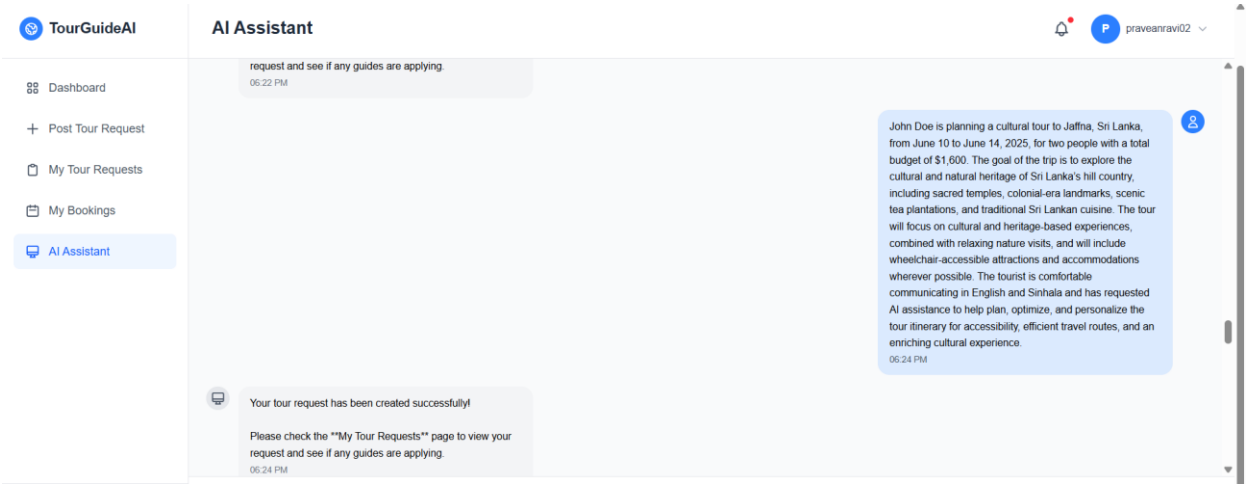
3. Tourist My Tour Requests



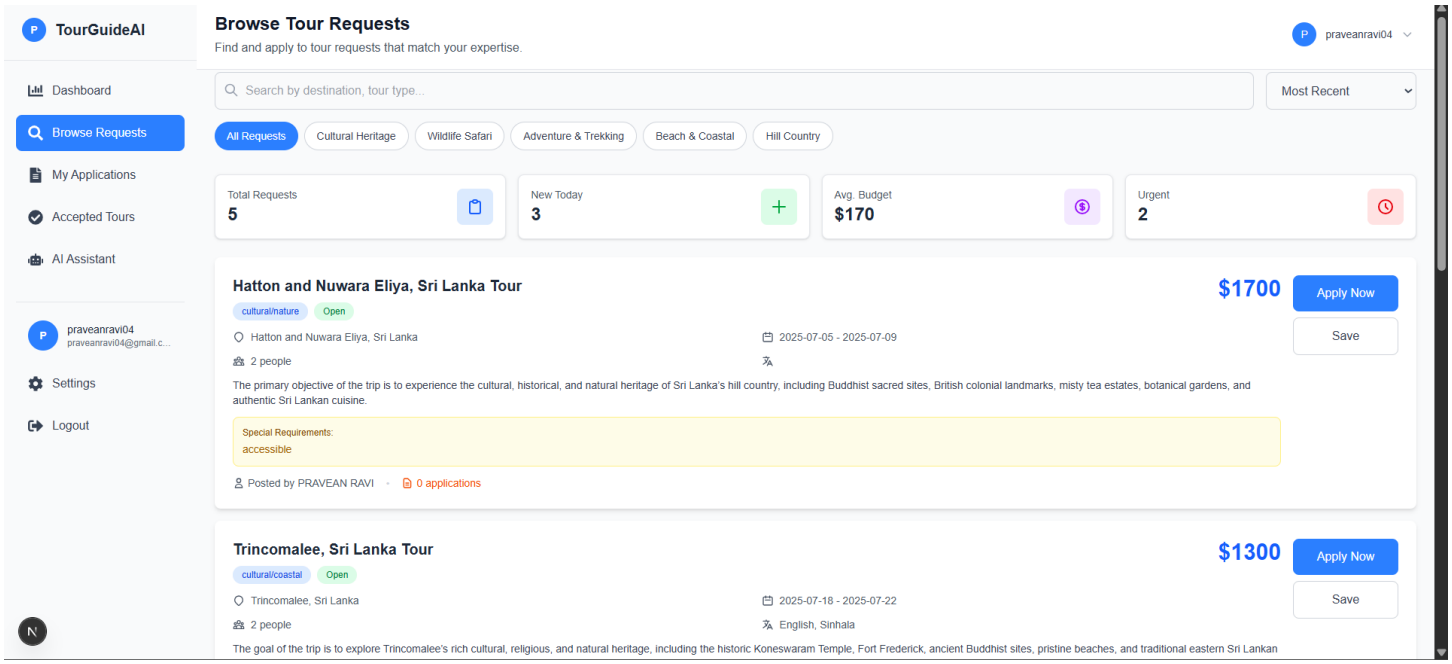
4. Tourist My Bookings



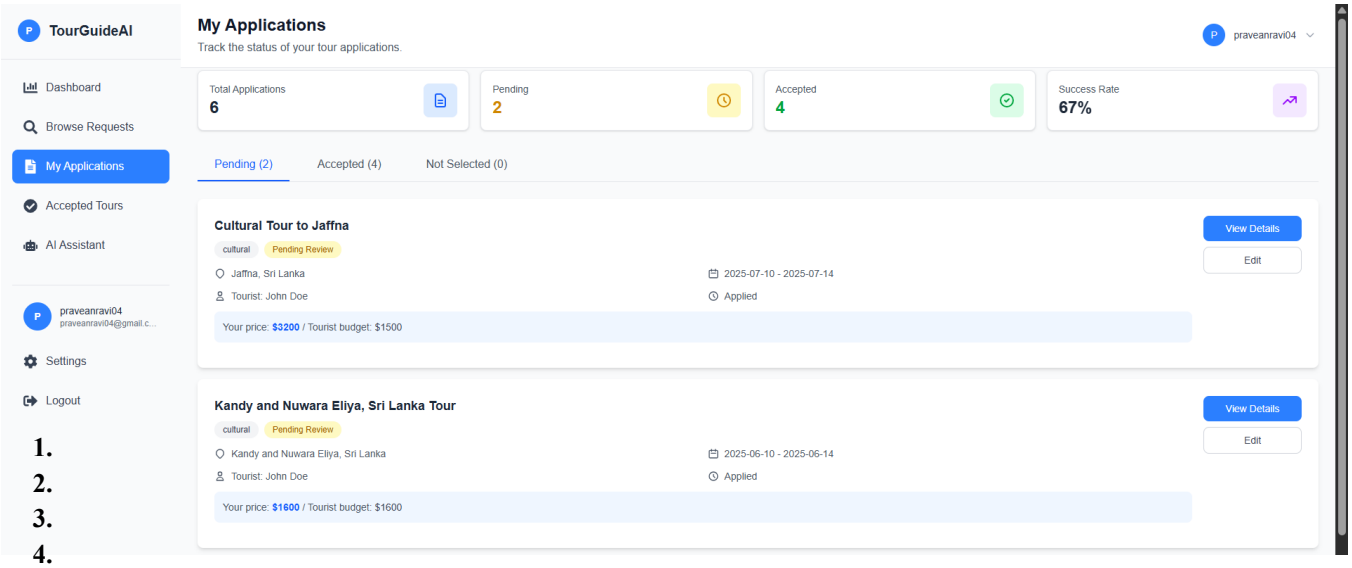
5. Tourist AI Assist



6. Guied Browse Tour Requests

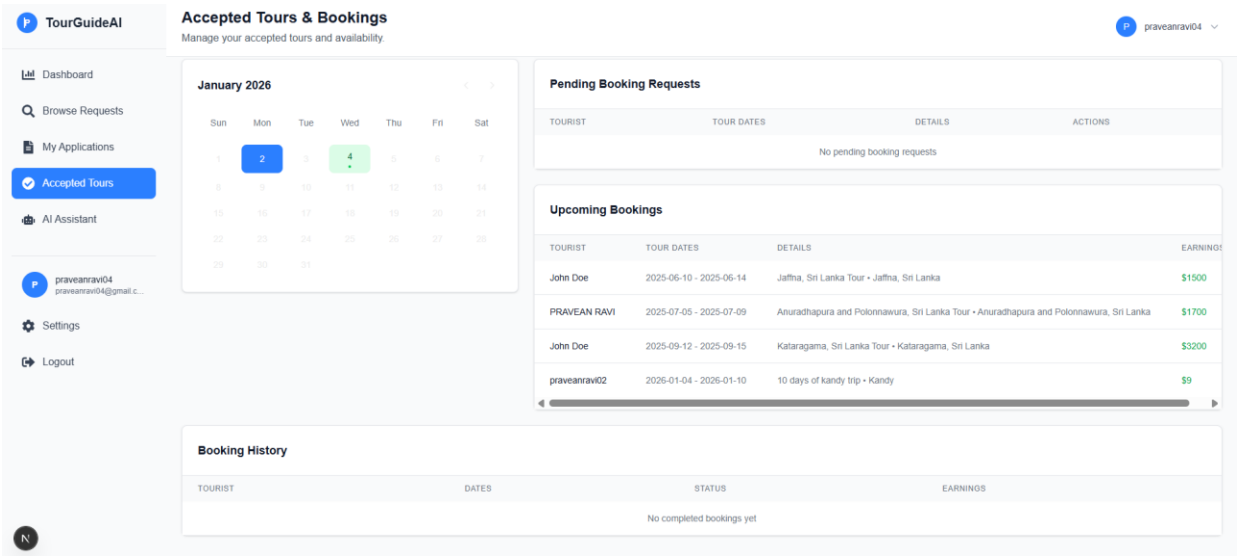


7. Guied My Applications

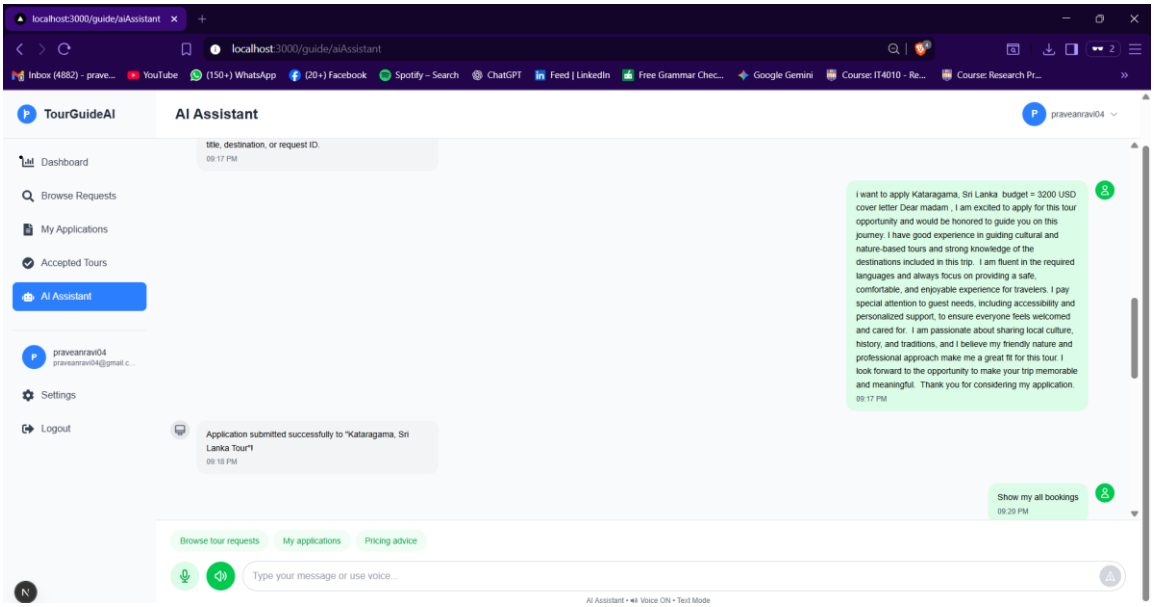


1.
2.
3.
4.

8. Guied Accepted Tours & Bookings



9. Guied AI Assistant



3.2 Hardware Interfaces

. The proposed system primarily interacts with standard hardware components including desktops, laptops, and mobile devices. The logical characteristics include compatibility with devices capable of running modern web browsers (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge). The physical characteristics involve interaction via input/output peripherals like keyboards, mouse, touchscreens, and network adapters. The system does not require specialized hardware. Communication between hardware and software is managed through HTTP/HTTPS protocols, ensuring secure and efficient data exchange.

3.3 Software Interfaces

Frontend Development

- React.js – Interactive and responsive user interfaces
- HTML5, CSS3, JavaScript Core web technologies
- Figma for prototyping

Backend Development

- Node.js with Express.js – Server-side logic and RESTful APIs
- PostgreSQL / MySQL – Relational database for booking and user data
- NoSQL (MongoDB) – Flexible storage for unstructured data

AI & Machine Learning

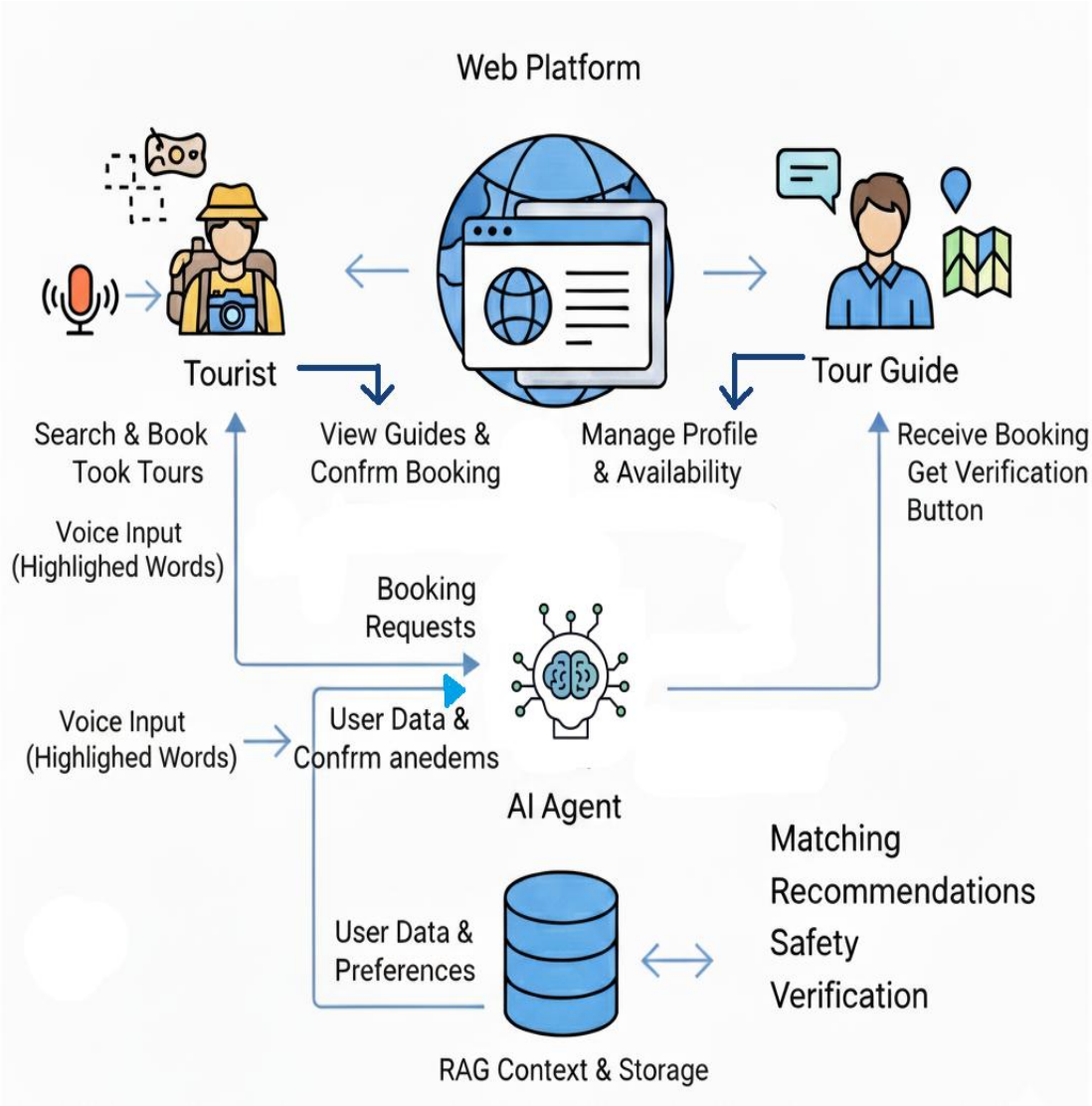
- RAG (Retrieval-Augmented Generation) – Personalized tourist–guide matching and recommendations
- LangChain / LlamaIndex – Orchestration of AI workflows
- OpenAI GPT / Hugging Face Models – NLP for Chat agent, keyword extraction, and multilingual support
- Vector Databases (Pinecone, Weaviate, FAISS) – Efficient retrieval for AI recommendations
- OCR Tools (Tesseract) – Document validation for guide verification

3.4 Communications Interfaces

Users access the application via a web browser (e.g., Chrome, Edge), connecting to the database and system backend. Communication is facilitated through the HTTP/HTTPS protocol for requests and responses, ensuring encrypted data exchange when necessary. The interface supports standard web interactions, including form submissions, data fetching, and dynamic content rendering. Server-client communication relies on API endpoints hosted by enabling seamless interaction between users and backend services.

4. System Features

The Software Solution Approach for the proposed AI-Powered Tour Guide Booking System with Voice and RAG emphasizes a systematic and iterative development process. It ensures that the final system is not only technologically robust but also user-friendly, secure, and aligned with the real-world needs of tourists and guides. The approach is divided into four main stages: requirement gathering, interactive modules and tools development, user experience and interface design, and testing and integration



4.1 Impact Analysis

4.1.1 Description and Priority

This features these stimulus–response sequences demonstrate how the system supports end-to-end tourist trip planning, from initial query to post-trip feedback, while integrating voice input, RAG-based AI intelligence, privacy controls, and trust mechanisms.

Priority: High

Feature	Benefit	Penalty	Cost	Risk	Priority Score
Tourist Trip Planning	9	7	5	4	7

Priority Score = (Benefit + Penalty) - (Cost + Risk)

4.1.2 Stimulus / Response Sequences – Tourist Trip Planning

Sequence 1: Tourist Initiates Trip Planning
The tourist accesses the system and initiates trip planning using voice or text input.

Response:
The system authenticates the user (if logged in), activates the AI Agent interface, and prepares to accept multilingual input (English, Sinhala, or Tamil).

Sequence 2: Voice/Text Query Submission
The tourist submits a trip-related query (e.g., “Plan a 3-day cultural trip in Kandy”).

Response:
The system processes the input, converts voice to text if necessary, and extracts highlighted keywords such as location, duration, interests, and budget.

Sequence 3: Context Retrieval Using RAG
The extracted keywords are passed to the retrieval module.

Response:
The system retrieves relevant documents from the vector database, including tourist attractions, cultural events, weather information, safety advisories, and available guides.

Sequence 4: AI-Based Trip Plan Generation
Retrieved contextual data is sent to the Large Language Model.

Response:
The AI Agent generates a personalized trip plan, including recommended places, suggested itineraries, and suitable tour guides based on preferences and verification status.

Sequence 5: Guide Matching and Presentation
The tourist requests to view recommended guides for the proposed trip.

Response:
The system displays a ranked list of verified and non-verified guides with limited profile details, ratings, availability, and estimated pricing.

4.1.3 Functional Requirements

REQ 1	The system shall allow tourists to submit trip planning queries using voice or text input in English, Sinhala, or Tamil.
REQ 2	The system shall process voice input by converting speech to text and extract highlighted keywords such as location, duration, interests, and budget.
REQ 3	The system shall retrieve relevant tourism-related information (attractions, guides, safety tips, events) from a vector database using RAG based on extracted keywords.
REQ 4	The system shall generate personalized trip plans and guide recommendations using a Large Language Model combined with retrieved contextual data.
REQ 6	The system shall allow tourists to view, select, and book verified or non-verified guides, while enforcing privacy controls that unlock personal details only after booking confirmation.

4.2.1 Guide Trip Acceptance Flow

This stimulus–response sequences describe how guides interact with the system to review, accept, modify, and complete trip plans, while maintaining privacy, security, and AI-assisted coordination throughout the booking lifecycle.

Priority: High

Feature	Benefit	Penalty	Cost	Risk	Priority Score
Guide Trip Acceptance Flow	9	7	4	4	5

Priority Score = (Benefit + Penalty) - (Cost + Risk)

4.2.2 Functional Requirements

Sequence 1: Guide Receives Trip Request
A tourist submits a trip plan and booking request to a selected guide.

Response:
The system notifies the guide through the dashboard and real-time notification system with summarized trip details.

Sequence 2: Guide Reviews Trip Plan
The guide opens the booking request to review itinerary details, dates, duration, location, and expected responsibilities.

Response:
The system displays the trip plan, tourist preferences, guide compensation details, and verification requirements while keeping sensitive tourist data masked.

Sequence 3: Availability Verification
The guide checks availability for the requested dates and time period.

Response:
The system validates the guide's calendar and flags any scheduling conflicts or confirms availability.

Sequence 4: Guide Accepts or Proposes Modification
The guide accepts the trip plan or proposes minor modifications (pricing, timing, or itinerary).

Response:
The system records the response and notifies the tourist with the updated plan or acceptance status.

Sequence 5: Tourist Confirmation
The tourist confirms the accepted or modified trip plan.

Response:
The system finalizes the booking, unlocks permitted personal details for both parties, and triggers optional payment processing.

Sequence 6: Pre-Trip Activation
The booking is confirmed and marked as active.

Response:
The system enables trip features such as emergency location tracking, in-app messaging, AI safety guidance, and itinerary reminders.

4.2.3 Functional Requirements

REQ 1:	The system shall notify the guide when a tourist submits a trip plan and booking request.
REQ 2:	The system shall allow the guide to view trip details, including itinerary, dates, duration, location, and proposed compensation, while masking sensitive tourist information prior to acceptance. device.
REQ 3:	The system shall validate the guide's availability against the requested trip schedule before allowing acceptance.
REQ 4:	The system shall allow the guide to accept the trip plan or propose modifications such as pricing, timing, or itinerary adjustments.
REQ 5:	Upon tourist confirmation, the system shall finalize the booking, unlock permitted personal details for both parties, and activate pre-trip features including messaging and emergency support.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR name	Description
Performance	• The system must maintain high responsiveness and stability during concurrent user sessions. • Voice and text queries should return AI-generated results within acceptable latency 3 to 5 seconds for text and 8 to 12 seconds for voice queries, including speech-to-text conversion and RAG retrieval. • The system should support at least 500 concurrent active users during peak load without service degradation. • Database queries, API responses, and vector searches must be optimized for efficiency using indexing and caching mechanisms. • System uptime must meet a minimum of availability, excluding scheduled maintenance periods.

5.2 Safety Requirements

NFR name	Description
Safety	• The application must ensure data and operational safety by implementing strict access control and backup mechanisms. • Personal and sensitive data (e.g., identity documents, GPS locations) must be encrypted both in transit and at rest. • Regular automated backups will safeguard data against accidental loss, while emergency features such as location sharing require explicit user consent. • The system should include safety notifications and fallback actions in case of communication or API failures. LLM responses must be filtered to prevent the generation of harmful, misleading, or culturally inappropriate content.

5.3 Security Requirements

NFR name	Description
Security	• The system must adhere to industry-standard security protocols to protect user privacy and prevent unauthorized access. • Communication between clients and servers must use HTTPS, and all sensitive data must be encrypted using AES-256. • Authentication and authorization shall follow OAuth 2.0 with role-based access control (RBAC) for tourists, guides, and administrators. • Security logging and monitoring must be implemented to track login attempts, API calls, and data access events. • The system should undergo periodic vulnerability scans and comply with secure software development lifecycle (SDLC) standards to prevent injection attacks, data breaches, and fraudulent activities.

5.4 Software Quality Attributes

- **Availability** - The system should be available 24 hours and handle high traffic volumes, especially during peak times.
- **Correctness** - The system must provide accurate information about identification, environmental hazards, and correct disposal methods & correct routes for collection centers.
- **Flexibility** - The system should allow for the addition of new features or content without disrupting existing functionality.
- **Maintainability** - The design of this system should be easy to maintain and easily modified as new requirements arise. It also doesn't take long to correct a mistake
- **Reliability** - The system should be reliable, functioning smoothly without errors
- **Testability** - The system's components must be independently testable, with clear and isolated responsibilities for each component to ensure correct behavior.
- **Usability** - The System interface must be user friendly and easy to understand.

5.5 Business Rules

Function	Tourist	Tour Guide	Admin	System Developer
Register / Create Account	✓	✓	✓	
Book Tour / Package	✓		✓	
Approve / Manage Tour Bookings		✓	✓	
Generate Reports / Analytics			✓	✓
Update / Maintain Tour Information		✓	✓	✓
Manage Users / Access Control			✓	
Integrate / Connect External Services			✓	✓

6. Other Requirements

Database Requirements

- **Database Scalability**: The database must be designed to handle increasing amounts of user data without affecting performance.
- **Data Backup and Recovery**: Automated data backup solutions must be implemented, along with a recovery plan to minimize downtime and potential data loss.

Legal Requirements

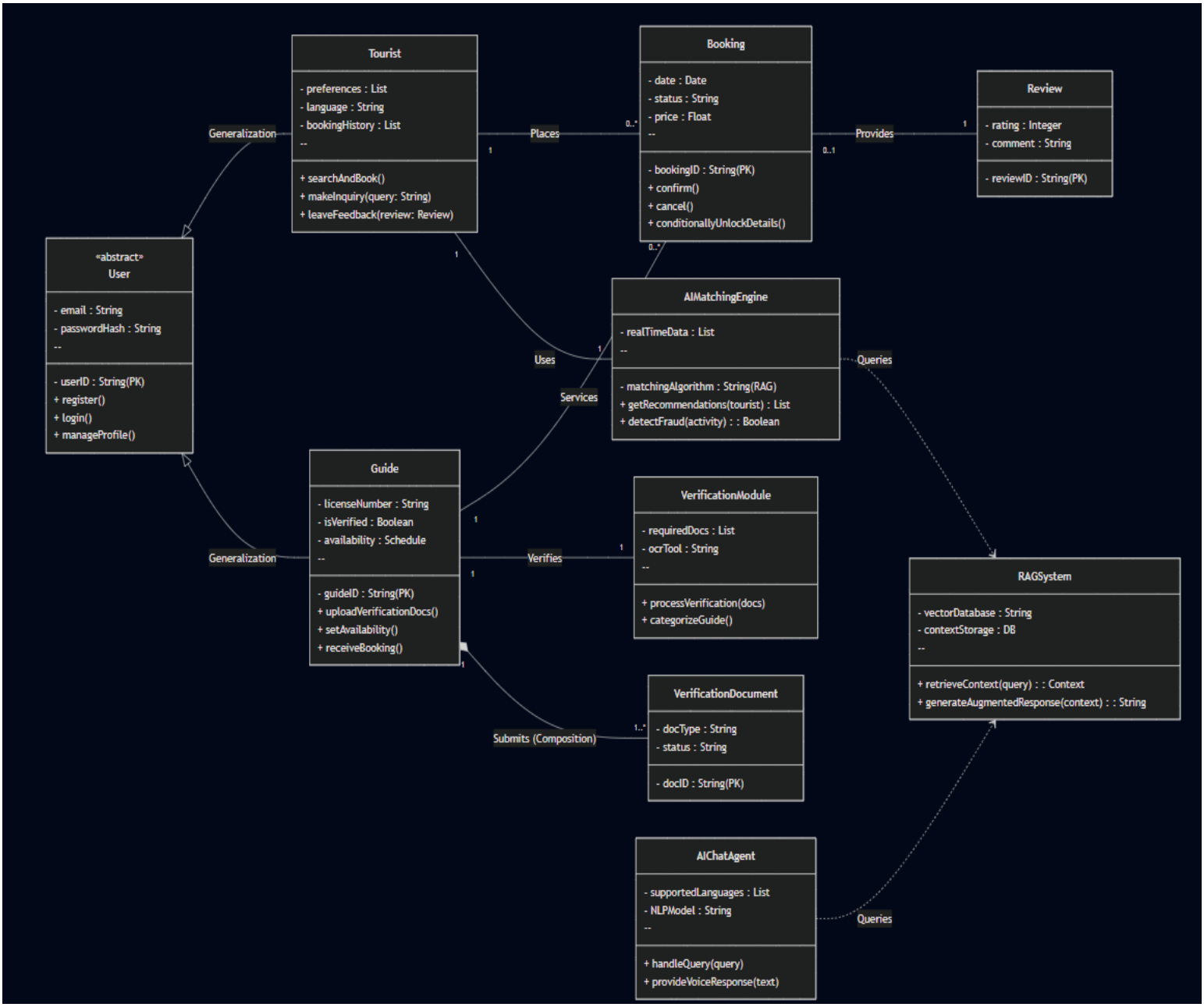
- **Terms of Service and Privacy Policy**: The System must present clear terms of service and a privacy policy to users during the registration process and within the system.
- **Compliance with Local Laws**: The System must comply with all relevant local regulations regarding data protection & the law of the country.

Appendix A: Glossary

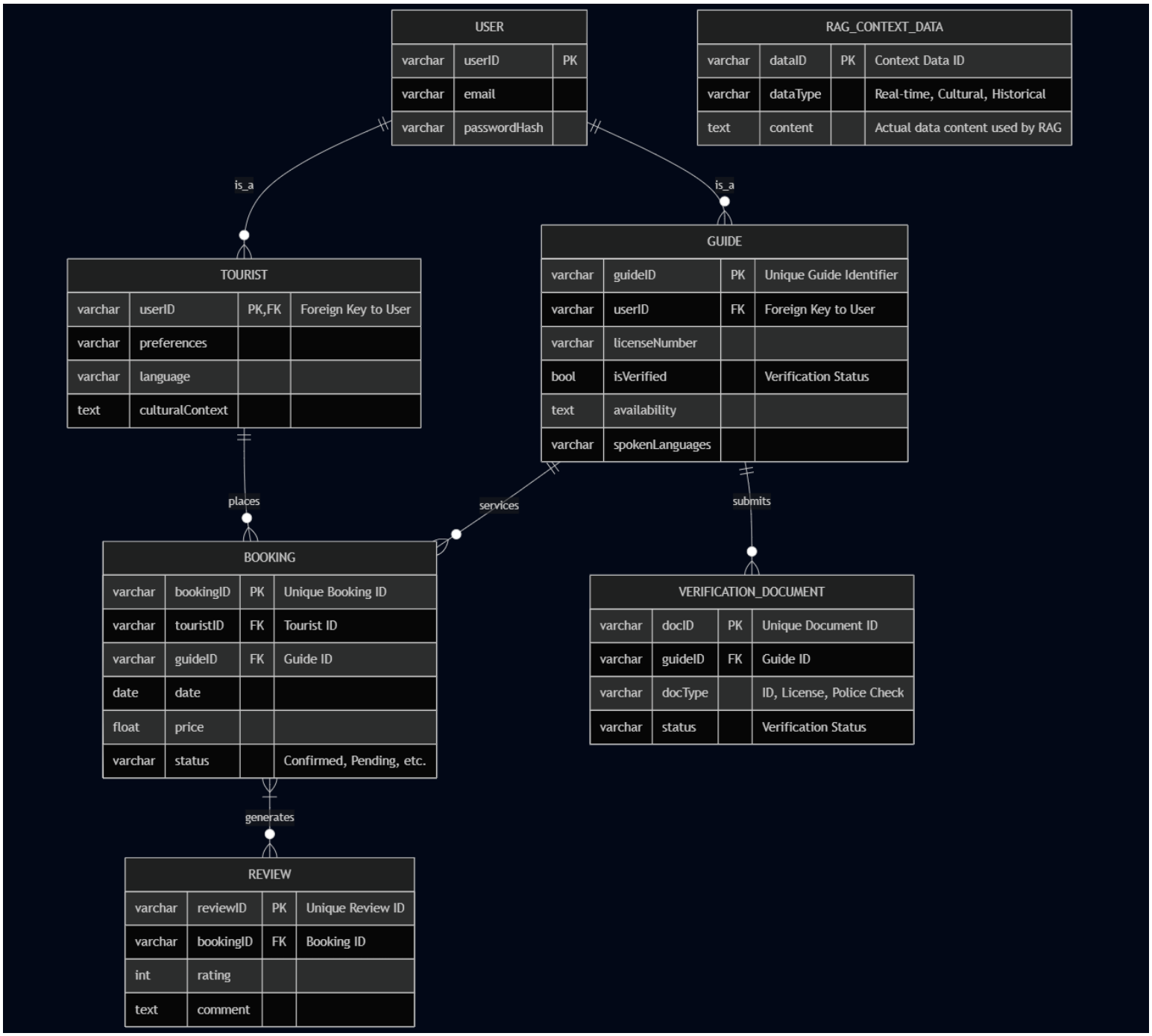
- **AI Agent**: *Intelligent component managing voice input, retrieval, and response generation.*
- **RAG (Retrieval-Augmented Generation):***AI technique combining document retrieval with language model responses.*
- **LLM (Large Language Model):** *Advanced AI model trained to understand and generate human-like text.*
- **Vector Database:** *Specialized database for storing and retrieving text embeddings efficiently.*
- **OCR (Optical Character Recognition): Technology** *that converts scanned or uploaded documents into machine-readable text.*
- **OAuth 2.0:** *Secure authorization framework for granting limited access to user data.*
- **API (Application Programming Interface):** *Set of rules that enable software systems to communicate with each other.*
- **GDPR:** *Data protection regulation ensuring privacy and security of personal information.*

Appendix B: Analysis Models

Class Diagram



Entity Relationship Diagram



Appendix C: To Be Determined List

Gantt *Chart*